

Homework 5

Due: Monday, April 28th

Credit: 150/100

Convolutional neural network. The standard interpretation of the elastic modulus of a rock sample is based on load-displacement data of nanoindentation, whose schematic is shown in Fig. 1. The unloading behavior correlates with the elastic modulus at the contact point, and the reduced modulus is expressed as follows:

$$E_r = \frac{\sqrt{\pi} S_{unloading}}{2 \sqrt{A}} \quad (1)$$

where E_r is the reduced modulus, $S_{unloading}$ is the slope of unloading, and A is the impression area. The reduced modulus controls the contact behavior and depends on the properties of the two media:

$$\frac{1}{E_r} = \frac{1 - \nu_{sample}}{E_{sample}} + \frac{1 - \nu_{indenter}}{E_{indenter}} \quad (2)$$

where ν_{sample} is the Poisson's ratio of the sample, E_{sample} is the elastic modulus of the sample, $\nu_{indenter}$ is the Poisson's ratio of the indenter, and $E_{indenter}$ is the elastic modulus of the indenter. The indenter's contribution to the reduced modulus is negligible because its elastic modulus is much higher than the elastic modulus of the sample; thus, the elastic modulus of the sample is determined from the unloading slope:

$$E_{sample} = (1 - \nu_{sample}) \left(\frac{\sqrt{\pi} S_{unloading}}{2 \sqrt{A}} \right) \quad (3)$$

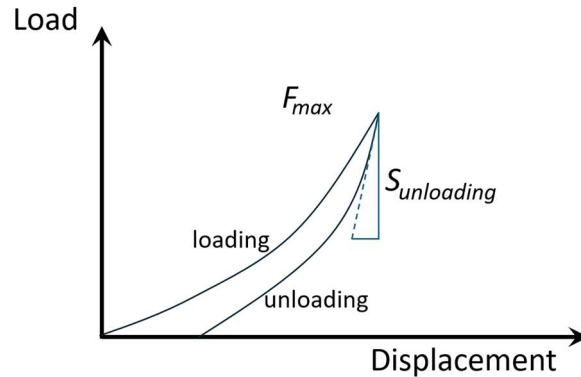


Fig. 1. Schematic of the load-displacement data.

This homework focuses on 1,000 tests. Each test includes an image, applied load, penetration depth, and elastic modulus. The applied load varied to generate micro-sized impressions of different sizes.

1. Augment the images ten times to increase their total amount to 10,000, which is a reasonable threshold in this problem, to improve deep learning performance. We practiced different augmentations in class, but you can also use other augmentations if you deem them necessary. Show augmented images of a single test of your choice to receive credit.
2. Use VGG16 with weights pretrained by ImageNet after replacing its top layer. Also, modify the loss function and activation function if needed. Train the model to determine the elastic modulus from the image. Show the scatter plot of the predicted versus actual elastic modulus of the test data, $E(\text{GPa})$.
3. *Bonus problem (50 points)*. Improve performance by using additional fully connected layers on top of your model and fine-tuning it. You may use another architecture or design your model from scratch. Keras provides various models that are easily accessible in Python: <https://keras.io/api/applications>